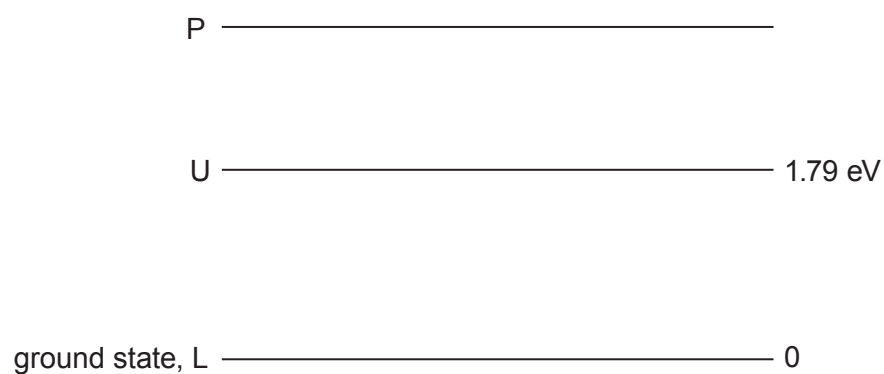


3. A simplified energy level diagram for a 3-level laser system is given.



- (a) (i) Calculate the wavelength of light emitted by stimulated emission. [3]

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- (ii) Assuming that some photons of this wavelength are already present in the laser cavity, explain why a population inversion is needed for light amplification to take place. [2]

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(b) The laser produces a light beam of power 6.0 mW.

(i) Show that the number of photons emitted per second is approximately  $2 \times 10^{16} \text{ s}^{-1}$ .  
[2]

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(ii) Calculate the momentum of the light leaving the laser per second. [2]

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(iii) Calculate the force exerted by the light beam on a perfectly reflecting surface, if it strikes the surface normally. [1]

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